Lahore University of Management Sciences
School of Science and Engineering, Department of Artificial Intelligence

AI651 – Deep Learning for Space, Time and Graphs

Fall 2026 • Assignment 2

Report

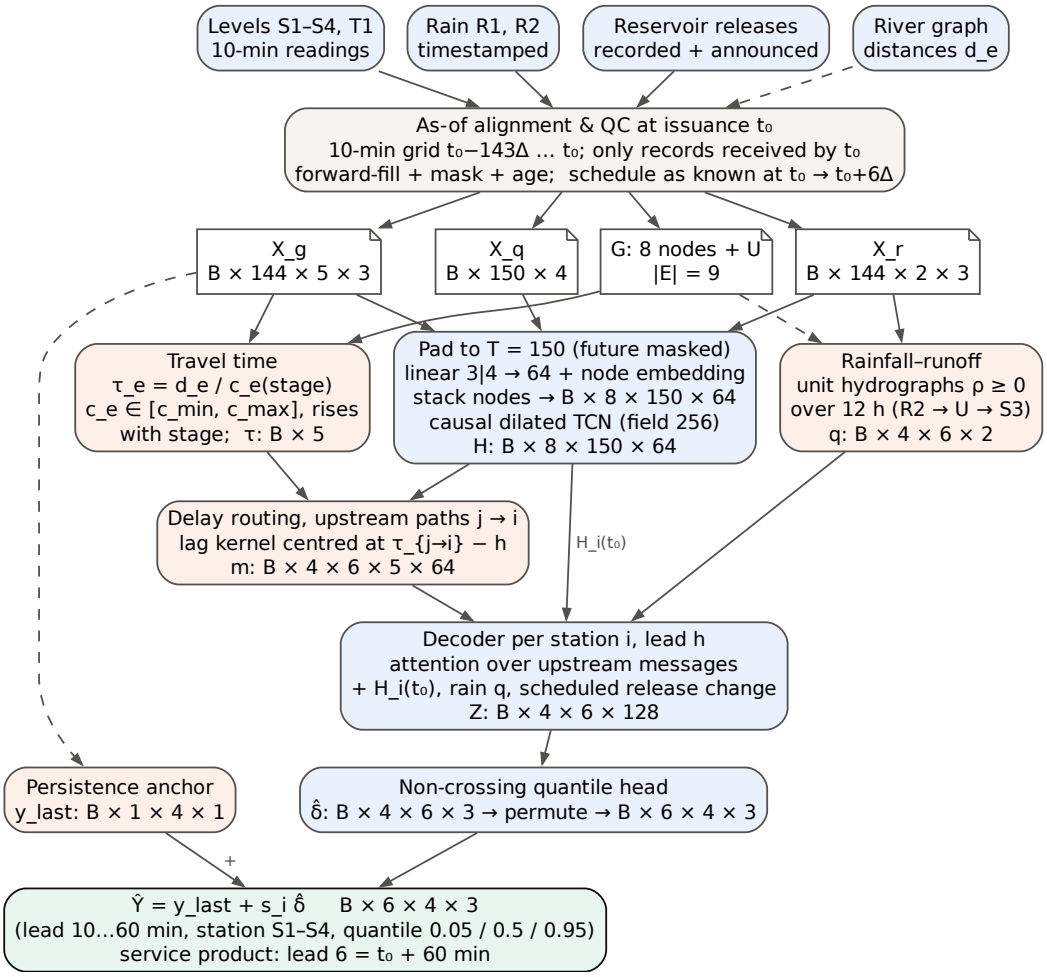
Student: Danyal Tayyub
Roll number: 24280023
Code repository: `https://github.com/danyaltayyub/DL4STG-PA2`

Contents. Question 1 proposes a river-level forecasting system under the six required headings. Question 2 trains a ConvLSTM forecaster on Moving MNIST, tests it under mirroring, rotation and speed-up, separates appearance, unfamiliar-motion and reach failures with models B–D, and builds a symmetrised final model. Code, GIFs and the pre-run predictions are in the repository (`Question2/`, `report/`). The appendix lists the use of generative AI.

1 Question 1 — River Forecasting Proposal

Proposal in one sentence. A small, physics-shaped spatio-temporal graph network forecasts the change in water level at S1–S4 over the next hour as quantiles. It routes upstream information along the river graph with stage-dependent travel times, feeds announced reservoir releases in as known future forcing, and adds the predicted change to the last observed level.

1.1 Architecture diagram



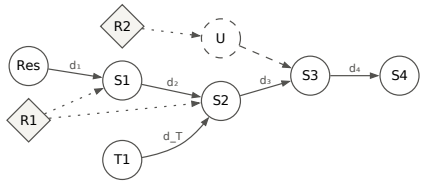


Figure Q1.1: Architecture (top) and the river graph used for routing (left). Blue blocks are learned networks; orange blocks have a fixed physical form with a few learned parameters; note-shaped boxes are the input tensors (axes are listed in the table of Section 1.3). Forecasts are issued at t_0, and $\Delta = 10$ min. In the graph, solid arrows are channel reaches of length d_e, the dashed node U is the unmonitored tributary (no data; its inflow is inferred), and dotted arrows attach the rain gauges to the catchments they represent.

1.2 Assumptions

Supplied facts. **F1**: 10-min water levels at S1–S4 and T1. **F2**: rainfall at R1 and R2 with timestamps and gauge locations. **F3**: recorded releases and schedules announced before issuance, with announcement and effective times. **F4**: station locations, connectivity and along-river distances. **F5**: travel times and the

unmonitored tributary's inflow are not supplied; future rainfall and future levels are unavailable at issuance. Everything below is an *assumption*, chosen to be the most defensible reading where the brief is silent.

ID	Assumption	Why it is reasonable	Effect on the design
A1	All timestamps use one clock (stored in UTC). Readings are nominally every 10 min with up to ±2 min jitter.	Standard for telemetered gauges.	Every reading is assigned to the grid slot $(t-\Delta, t]$ that contains it (Section 1.3).
A2	Most records arrive within 5 min, but outages can delay them by hours. Each record keeps its *arrival* time.	Typical of radio/cellular telemetry.	Inputs are assembled *as of* t_0 by arrival time, and a data-age feature marks staleness. Training replays arrival times (if they are not logged, see request I5).
A3	At least 3 years of overlapping records exist, with several high-flow events and an archive of schedules *as announced*.	The minimum for learning seasonal and event behaviour.	A small model (about 2×10^5 parameters) with physical structure. Train/validation/test split chronologically by year, with flood events in every split.
A4	Level sensors are accurate to ±1–2 cm, with occasional spikes, flat-lines and logged datum shifts. Rain gauges are tipping buckets (0.2 mm) that undercatch in wind.	Typical gauge specifications.	Range and rate-of-change checks turn bad values into missing values. The sensor noise sets the floor for useful accuracy.
A5	Between stations, water only flows downstream; backwater at the confluences is negligible.	Steep mountain river (see the illustration).	Directed graph: a station only uses data from upstream nodes.
A6	Travel time depends on the water level: higher water gives a faster flood wave (in-bank kinematic-wave behaviour), with mean velocities between 0.3 and 4 m/s.	Standard open-channel hydraulics.	Each edge's wave speed c_e is bounded and increases with stage, so its travel time is d_e/c_e.
A7	Releases enter just above S1, and a release change reaches S1 within one hour.	S1 is directly below the dam in the map.	Scheduled releases inside the forecast hour act directly on S1, and on downstream stations when their path delay is short.
A8	The unmonitored tributary is driven mainly by rain on its own catchment, which R2 (the nearest gauge) represents best; its baseflow varies slowly.	Gauge proximity in the map; baseflow changes over days.	Pathway R2 $\to$ U $\to$ S3 through a rainfall–runoff kernel. Slow baseflow is absorbed by S3's own history.
A9	R1 represents rain on the hillslopes between the dam and S2. T1 sits upstream of its confluence.	Gauge positions in the map.	R1 connects to every station from S1 down; T1 reaches S2 through edge T1→S2 with its own delay.
A10	Rain affects main-river levels after lags of 30 min to 12 h. The total reservoir-to-S4 travel time is under 12 h. Snowmelt follows a daily cycle.	Small mountain catchment with snow (illustration).	A 24 h history (144 steps) covers the longest lag and one full daily cycle. The encoder's receptive field is 256 steps.
A11	Gaps can cluster during storms, because power and telemetry fail when flows are high.	Common in flood operations.	Training simulates outages that are concentrated on high-flow periods (Section 1.5).
A12	Channel geometry, gauge datums and the reservoir's operating policy are stable over the record; slow drift is acceptable.	Needed to learn from history at all.	Periodic retraining plus drift monitoring; out-of-range stages are flagged.

1.3 Forecasting task and data specification

Issuance and horizon. A forecast is issued every 10 min at grid times t_0 (`hh:m0`), as soon as the inputs have been assembled, typically within a minute of t_0. The **history window** is $L = 144$ steps (24 h), $t_0-143\Delta, \ldots, t_0$, long enough for A10. The model predicts all six leads $t_0 + h\Delta$, $h = 1, \ldots, 6$, jointly. This costs nothing extra, gives the intermediate leads as supervision, and yields a trajectory for operators. The **service product** is $h = 6$, the level at $t_0 + 60$ min at S1–S4, given as a median and a 90% interval (quantiles 0.05, 0.5, 0.95).

Tensor	Shape	Axes and content	Timestamps / time coverage
X_g	$B \times 144 \times 5 \times 3$	batch; history step; gauge (S1, S2, S3, S4, T1); [level as a per-station z-score, observed mask, data age in hours]	Slots $t_0 - 143\Delta, \ldots, t_0$. Slot t holds the latest reading timed in $(t - \Delta, t]$ that had arrived by t_0.
X_r	$B \times 144 \times 2 \times 3$	batch; history step; rain gauge (R1, R2); [$\log(1 + \text{mm})$ summed over the slot, mask, age]	Same slots. Future rain is unknown, so there are no future slots.
X_q	$B \times 150 \times 4$	batch; 144 past + 6 future steps; [recorded release (scaled by capacity), scheduled release as known at t_0, recorded-value mask, is-future flag]	$t_0 - 143\Delta, \ldots, t_0 + 6\Delta$. Past slots hold the actual and the then-scheduled rate; future slots hold the schedule from the latest announcement made by t_0.
G	8 data nodes + latent U; $\lvert E\rvert = 9$	directed edges with lengths d_e (km); node types (gauge, rain, reservoir)	Static.
y^{last}	$B \times 1 \times 4 \times 1$	batch; –; S1–S4; last valid level in metres	The time of the last valid reading, which is $\le t_0$ (its age is in X_g).
$\hat{Y}$ (output)	$B \times 6 \times 4 \times 3$	batch; lead $h = 1..6$; S1–S4; quantile (0.05, 0.5, 0.95); levels in metres above station datum	$t_0 + 10, \ldots, t_0 + 60$ min; served at $h = 6$.
Y, M_Y (training only)	$B \times 6 \times 4$	observed level at each lead, and a mask for missing targets	$t_0 + h\Delta$, $h = 1, \ldots, 6$.

Batch axis. During training, B is a set of issuance times sampled from the training years (e.g. $B = 256$). In operation $B = 1$, or $B = 2$ when a release scenario is run alongside (Section 1.5).

Alignment across locations and sampling times. All sources are put on one 10-min grid that ends at t_0:

- **Levels** snap to their slot (A1). An empty slot is forward-filled from the last valid value, with mask 0, and its age keeps growing.
- **Rain** accumulations are summed per slot.
- **Releases** are step functions sampled at slot times.
- **Schedules** are *as-of joined*: for each future slot, use the rate from the latest announcement whose announcement time is $\le t_0$.

Training uses the same rule, so the model never sees a schedule, reading or rain value it would not have had at t_0. Uneven station spacing is not resampled away. It enters through the edge lengths d_e and the travel-time module, so each station is aligned with the *lagged* state of its upstream nodes, not with their same-time state. To combine sources in one encoder, gauge and rain inputs are padded with six masked future slots (a common axis, $T = 150$), projected per node type to 64 channels and stacked along the node axis: $B \times 8 \times 150 \times 64$.

Information available at issuance. Everything that had arrived by t_0: levels and rain up to t_0, releases recorded up to t_0 (possibly lagging), schedules announced up to t_0 (covering the future), and the static graph. *Not* available: future rain, future levels, schedule changes not yet announced, and the unmonitored tributary's inflow.

1.4 Model computations and design justification

Notation. $\Delta = 10\,\text{min}$, issuance time t_0, lead $h \in \{1, \ldots, 6\}$, hidden width $d = 64$. Nodes: reservoir Q, gauges S_1–S_4 and T_1, rain gauges R_1, R_2, and latent U (no data). $\text{Up}(i)$ is the set of gauges and reservoir upstream of station i; the graph is a tree, so the path $j \to i$ is unique. σ is the logistic function, and sp is softplus.

(1) Projection and encoder. For node j of type $\kappa(j)$ and slot t: $e_j(t) = W^{\kappa(j)} x_j(t) + p_j$, where p_j is a learned node embedding. Then 8 residual causal dilated convolution layers (kernel 2, dilation 2^ℓ) are applied:

$$H_j^{(\ell+1)}(t) = H_j^{(\ell)}(t) + \text{GELU}\big(W_0^{(\ell)} H_j^{(\ell)}(t) + W_1^{(\ell)} H_j^{(\ell)}(t - 2^\ell\Delta)\big), \qquad H_j = H_j^{(8)} \in \mathbb{R}^{150 \times d}.$$

Causality means $H_j(t)$ only depends on inputs at slots $\le t$. In the future slots, the reservoir carries the announced schedule, and gauges and rain carry only "unobserved" tokens.

(2) Travel time (physical form, learned parameters). For edge $e = (j \to k)$, with $\bar{y}_j$ and s_j the training

mean and standard deviation of station j's level:

$$c_e = c_e^{\min} + (c_e^{\max} - c_e^{\min})\,\sigma\big(\mathrm{sp}(\alpha_e)\,(y_j^{\mathrm{last}} - \bar{y}_j)/s_j + \beta_e\big), \qquad \tau_e = \frac{d_e}{c_e\,\Delta}, \qquad \tau_{j\to i} = \sum_{e\in \mathrm{path}(j,i)} \tau_e.$$

c_e is bounded (A6) and increases with stage, because $\mathrm{sp}(\cdot) \geq 0$. The travel times τ are measured in steps and can be read off for checking against hydrological knowledge.

(3) Delay routing. Upstream water that reaches station i at $t_0 + h\Delta$ passed node j about $\tau_{j\to i}$ steps earlier, i.e. $k^\star = \tau_{j\to i} - h$ steps before t_0. Let $k \in \{-6, \ldots, 143\}$ count steps before t_0 (negative values are future slots). Then:

$$\kappa_{ji,h}(k) = \frac{\exp\big(-(k-k^\star)^2/2\varsigma_{ji}^2\big)}{\sum_{k'}\exp\big(-(k'-k^\star)^2/2\varsigma_{ji}^2\big)}, \qquad m_{ji,h} = \sum_k \kappa_{ji,h}(k)\,H_j(t_0 - k\Delta), \qquad \phi_{ji,h} = \sum_{k<0}\kappa_{ji,h}(k).$$

$\varsigma_{ji} = \mathrm{sp}(\cdot)$ is a learned spread that represents wave dispersion (Muskingum-like attenuation). $\phi_{ji,h}$ is the share of the kernel that falls in the future. For the reservoir, those slots hold the *announced schedule*, so a release change reaches each station after its physical delay. For a gauge they hold no data, so ϕ tells the decoder that this upstream signal has not arrived yet; the decoder then leans on nodes further upstream, whose delays are longer.

(4) Rainfall–runoff. Each rain gauge r connects to every station downstream of where it attaches (R1: S1–S4; R2, through U: S3, S4), using a non-negative 12 h unit hydrograph:

$$q_{ri,h} = \sum_{\ell=0}^{71} \rho_{ri}(\ell)\,R_r\big(t_0 + h\Delta - \ell\Delta\big), \qquad \rho_{ri}(\ell) = \mathrm{sp}(\theta_{ri\ell}) \geq 0, \qquad R_r(t > t_0) = 0.$$

(5) Decoder and forecast mapping. Query $u_{i,h} = W_q[H_i(t_0); p_i; \eta_h]$, where η_h is a lead embedding. Each upstream source contributes the key $k_{ji,h} = W_k[m_{ji,h}; \tau_{j\to i}; \phi_{ji,h}]$:

$$a_{ji,h} = \underset{j\in \mathrm{Up}(i)}{\mathrm{softmax}}\big(u_{i,h}^\top k_{ji,h}/\sqrt{d}\big), \qquad z_{i,h} = \mathrm{MLP}\big([u_{i,h}; \textstyle\sum_j a_{ji,h} W_v k_{ji,h}; q_{\cdot i,h}; \delta_h^Q]\big) \in \mathbb{R}^{128},$$

where δ_h^Q is the scheduled change in release between t_0 and $t_0 + h\Delta$. The head $(o_1, o_2, o_3) = W_o z_{i,h}$ gives quantiles of the standardised change that cannot cross:

$$\hat{\delta}^{0.5} = o_1, \quad \hat{\delta}^{0.05} = o_1 - \mathrm{sp}(o_2), \quad \hat{\delta}^{0.95} = o_1 + \mathrm{sp}(o_3), \qquad \hat{Y}_{h,i}^{(q)} = y_i^{\mathrm{last}} + s_i\,\hat{\delta}_{i,h}^{q}.$$

(6) Training objective. A masked pinball loss, weighted towards the service lead ($w_6 = 2$, $w_{1..5} = 1$), plus a smoothness penalty on the unit hydrographs:

$$\mathcal{L} = \frac{\sum_{b,h,i} M_{Y,bhi}\,w_h \sum_q \rho_q\big(Y_{bhi} - \hat{Y}_{bhi}^{(q)}\big)}{\sum_{b,h,i} M_{Y,bhi}\,w_h} + \lambda \sum_{r,i,\ell}\big(\rho_{ri}(\ell{+}1) - 2\rho_{ri}(\ell) + \rho_{ri}(\ell{-}1)\big)^2, \qquad \rho_q(u) = \max(qu, (q{-}1)u).$$

Training uses AdamW with early stopping on the validation pinball loss at $h = 6$, and random input masking (Section 1.5). Interval widths are then calibrated per station on the validation years (split-conformal adjustment), so the 90% band covers about 90%.

Why these choices.

- *Predict the change, anchored on the last level.* Over one hour, persistence is the strongest simple forecast. Learning the change removes datum differences between stations and leaves the network the dynamics.
- *Explicit travel-time routing instead of stacked message passing.* With K message-passing layers, information only travels K hops, and it must also arrive at the right *lag*. On a 4-hop chain, routing along each upstream path gives full reach in hops and in time in one step, with a handful of parameters per edge. Travel times that change with conditions (A6) are modelled directly instead of being averaged away.
- *Causal dilated convolutions instead of temporal attention.* A 256-step receptive field costs linear time and memory. Attention would cost $O(L^2)$, and its main benefit, finding the relevant lag, is already provided by the routing kernel.
- *Non-negative unit hydrographs.* More rain can never lower the forecast, and a linear kernel can be learned from few storms.
- *Quantile outputs.* Flood preparedness needs risk, e.g. $P(\mathrm{S4} > \mathrm{threshold})$, not just a point value.

How each driver reaches each forecast.

- **S1**: its own history; the reservoir's recorded and announced releases, with a short delay (announced changes within the hour matter most); rain at R1.
- **S2**: adds S1 and T1, each through its own delay; the reservoir through the path $Q \to$ S1 $\to$ S2; R1.
- **S3**: adds S2 and everything upstream of it. The *unmonitored tributary* enters in two ways: R2's rainfall–runoff kernel, and S3's own history, whose departure from routed S2 is a running proxy for the tributary's inflow.
- **S4** (next to the town): all of the above through S3 and the longer paths. For short leads, S3's recent state dominates.

Physical assumptions versus learned mappings.

- *Physical, fixed by design:* graph topology and flow direction, edge lengths, travel time = distance / wave speed, wave-speed bounds that increase with stage, causal and as-of masking, non-negative rain response, the persistence anchor, and announced releases entering only through path delays.
- *Learned:* the wave-speed curves, dispersion, unit hydrographs, encoder, attention, decoder and quantiles.

A separate hydraulic simulator is not needed: the physics enters through structure and constraints. A simulator's output could later be added as one more input node (request I2).

1.5 Operation and exceptional conditions

(i) Missing observations.

- Missing or quality-rejected values are forward-filled with mask 0 and a growing age, never set to zero, so input scales stay meaningful.
- If a station's own reading at t_0 is missing, y^{last} is its last valid value, and the age feature lets the network widen its intervals and rely on upstream nodes.
- During training, inputs are randomly masked: single slots, runs of up to 6 h, and whole-station outages concentrated on high-flow periods (A11). Missing targets are excluded through M_Y.
- A station whose data are older than 6 h is flagged "degraded" on the dashboard. Its forecast is still issued, from upstream routing plus its last value.

(ii) Delayed observations.

- Inputs are assembled by *arrival* time (A2). A late reading is invisible to the forecasts issued before it arrives, and is used automatically from the next issuance on.
- The model keeps no state between runs (each forecast is recomputed from the 24 h window), so late data never corrupt a carried-over state.
- Training replays the logged arrival times, so the network sees the same kind of staleness it meets in operation.

(iii) Announced future releases.

- These are *known future inputs*. Each forecast uses the latest announcement made by t_0, as-of joined onto the future slots of X_q, and they act through the routing delays ($\phi > 0$ for the reservoir).
- If no announcement covers the hour, the current rate is assumed to continue, and a flag is set.
- Announcements made *after* t_0 are never used, either in training or in operation.

(iv) Differences between scheduled and actual releases.

- Past slots of X_q hold both the recorded and the then-scheduled rate. The network therefore learns how reliably schedules are followed, and widens its intervals when recent deviations are large.
- Once a deviation is recorded, it replaces the schedule for those past slots at the next issuance.
- For operators, a second scenario is run in the same batch ($B = 2$: "as scheduled" and "current rate continues"). The spread between the two shows how sensitive each station is to the release decision.

Feasibility every 10 minutes. The model has about 2×10^5 parameters. One forecast is a single forward pass over 8×150 node-steps, on the order of 10^7 multiply-adds: milliseconds on one CPU core. Assembling 24 h of records is a few small database queries. A run therefore takes seconds, far inside the 10-minute cycle, which leaves room for scenario runs, an ensemble of seeds and retries. If the model service fails, persistence plus routed upstream change is issued as a fallback and labelled as such.

1.6 Uncertainty, limitations, and additional information

Item	Forecast impact	Response, or what remains unresolved
Unmonitored tributary	Its inflow is invisible. This mainly affects S3 and S4 during local storms on its catchment.	Inferred from R2's kernel and S3's departure from routed S2, with wider quantiles when R2 is raining. *Unresolved:* storms that R2 misses, and the size of the inflow from rain falling within the hour.
Changing travel times	Wrong timing of a flood wave gives the largest errors on the rising limb.	Wave speed depends on stage, within physical bounds, and travel times are checked against observed event lags. *Unresolved:* over-bank flow at record floods can slow the wave, outside the training range; such stages are flagged.
No future rainfall	Rain falling in the next hour on hillslopes near S1/S2 is not in the inputs, so rapid rises are under-forecast.	Quantiles widen with current rain intensity. Fully fixed only by nowcasts (I3).
Release deviations	An unannounced change shows up first at S1 and then propagates downstream.	Deviation features and the two-scenario output (Section 1.5).
Data gaps in storms	Fewer inputs exactly when accuracy matters most.	Masking-robust training, degraded flags and persistence fallback. Remaining risk: large gaps at S3/S4.
Rare extremes	Few large floods to learn from, so the highest stages extrapolate.	Physical structure, a small model, held-out events for testing, and conformal calibration. Not fully resolvable without longer records.
Levels instead of discharge	Inflows do not add up in level units, so no mass balance can be imposed.	The level response is learned (non-linear decoder). Resolved by rating curves (I1).
Sensor drift / datum shifts	Bias at one station that propagates downstream.	Quality-control rules, a residual monitor against routed upstream levels, and retraining after logged datum changes.
Snowmelt and non-stationarity	Daily melt cycles and changing operating policies shift the input–output relation.	The 24 h window includes yesterday's cycle; retraining plus drift monitoring. Temperature data would help (I6).

Additional information requests. The system works without every one of them; each would improve it.

- **I1. Rating curves at S1–S4 and T1.** These let the model work in discharge, impose mass balance, and estimate the unmonitored tributary's inflow as Q_{S3} − routed Q_{S2} − routed Q_{T1}.
- **I2. Channel cross-sections and slopes, past travel-time studies, or a hydraulic model.** These tighten the wave-speed bounds and priors, and a hydraulic model's output could be added as an input node.
- **I3. Rainfall nowcasts (radar or weather-model forecasts) for the next 1–3 h.** These cover the largest missing driver; they would enter as future slots of X_r.
- **I4. A temporary gauge on the unmonitored tributary, even for one season.** It would let us calibrate the R2 $\rightarrow U$ pathway and test the inflow proxy.
- **I5. Arrival-time logs, quality-control flags and sensor maintenance logs.** These make the replay of delays and gaps in training realistic. Without them, worst-case latencies are assumed.
- **I6. Air temperature and snow cover, plus the schedule archive with announcement times and the reservoir operating rules.** These capture snowmelt and how reliably schedules are followed. If the archive lacks announcement times, schedules cannot be reconstructed as of t_0; training then falls back to recorded releases with added noise.
- **I7. Flood thresholds at S4 (town) and the cost of misses versus false alarms.** These set the loss weights and warning thresholds, and let the quantile output be turned into exceedance probabilities.

2 Question 2 — Motion or Memory?

Tables and figures are numbered as in the handout. Code is in `Question2/` (`SPEC.md` restates the protocol; `checks.py`, `train.py`, `evaluate.py`). Every learned-model entry is *mean (seed 0, seed 1)*. Errors are given to three significant figures, with scientific notation below 10^{-3}. Unless stated otherwise, results are on the 1,000 test videos at 1× with $t = 1$. Hardware: NVIDIA GeForce RTX 4060 Laptop GPU (8 GB), PyTorch 2.9.0 (CUDA 12.8), float32.

2.1 Setup

Table 1: Sanity checks of the transformations and window construction (the "Yours" column is computed by `checks.py`).

Row	Check	Expected	Yours
(a)	$\max \lvert M(M(x)) - x \rvert$	0, `torch.equal`: True	0, `torch.equal`: True
(b)	$\max \lvert R^4(x) - x \rvert$	0, `torch.equal`: True	0, `torch.equal`: True
(c)	$\max \lvert MRM(x) - R^{-1}(x) \rvert$	0, `torch.equal`: True	0, `torch.equal`: True
(d)	bright pixel of $M(p)$ (frame, row, col.)	(7, 10, 43)	(7, 10, 43)
(e)	bright pixel of $R(p)$ (frame, row, col.)	(7, 43, 10)	(7, 43, 10)
(f)	1× input / target frames	1–5 / 6–10	1–5 / 6–10
(g)	2× input / target frames	1,3,5,7,9 / 11,13,15,17,19	1,3,5,7,9 / 11,13,15,17,19
(h)	3× input / target frames	1,4,7,10,13 / 16,19	1,4,7,10,13 / 16,19

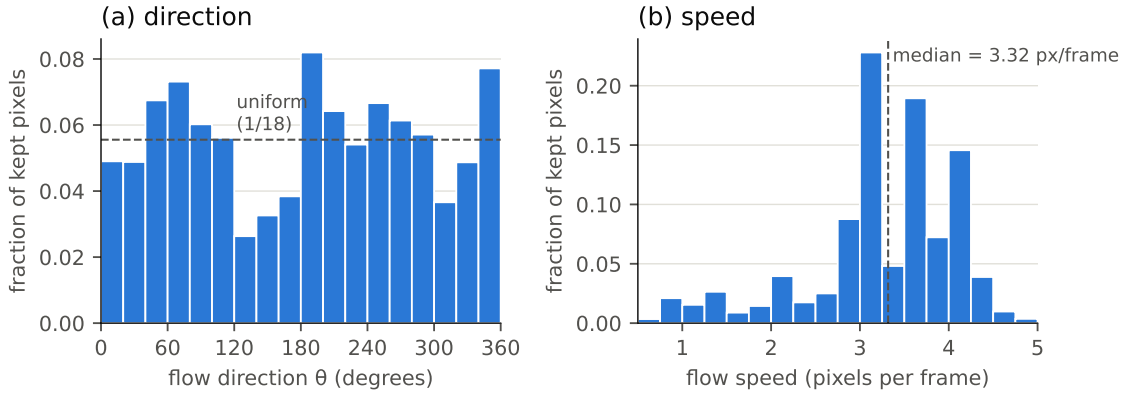


Figure 1: Optical flow on the first 50 training videos (950 consecutive-frame pairs, Farneback). Pixels kept: 103,050 (above 0.2 intensity in both frames and flow magnitude above 0.5 px). (a) Flow direction, $\theta = \text{atan2}(-d_y, d_x)$, as a fraction of kept pixels; the dashed line is the uniform level 1/18. (b) Flow speed; median 3.32 px/frame (dashed). Bars beyond 5 px/frame (0.46% of kept pixels) are not drawn.

Q2.1. *Mirrored and rotated motion is already familiar.*

- Figure 1(a) has training motion in every direction and no empty sector. The bins vary between 0.026 and 0.082 around the uniform 0.056. That variation is what about 100 digit trajectories in 50 videos produce, not a preferred direction.
- Mirroring maps $\theta \mapsto 180° - \theta$, and a 90° rotation maps $\theta \mapsto \theta + 90°$. Applied to every training video, either would give a histogram that differs from Figure 1(a) only by this sampling noise. So a model trained on these data has already seen mirrored and rotated *trajectories*, while mirrored and rotated *digits* remain unfamiliar.

Faster motion is not.

- Speeds are concentrated: median 3.32 px/frame, interquartile range 3.0–3.9, 90th percentile 4.1 (Figure 1(b)).
- 2× and 3× correspond to about 6.6 and 10.0 px/frame. Both lie outside what the training data contain.

Overlap bias. Where digits overlap, Farneback's single smooth flow field averages two motions. This produces short or wrongly directed vectors, which feed the low-speed tail below 2 px/frame or are dropped by the 0.5 px

threshold. Two 28×28 digits in a 64×64 frame overlap in only a minority of frames. The bias can therefore blur the speed histogram's lower tail and add noise to direction bins, but it cannot fill or empty a whole direction sector, nor move the 3–4 px/frame peak. Neither conclusion changes.

Q2.2. Each transformation keeps the videos physically valid, because constant velocity with elastic bounces in a square box is preserved:

- **Mirror** leaves the motion statistics essentially unchanged (Q2.1) and changes appearance: a mirrored digit is not a digit.
- **Rotation** likewise leaves the motion unchanged and changes appearance more: strokes swap orientation.
- **Speed-up** leaves appearance unchanged (the same frames, subsampled) and changes only the motion, pushing it outside the training range.

Each test therefore moves one factor while holding the other fixed. A failure under mirror or rotation points to appearance; a failure under speed-up points to motion. Mirror and rotation give two doses of appearance change. A model that has learned the motion rule itself should pass all three.

2.2 Baseline: ConvLSTM

Table 2: Test MSE per horizon, $E^{(5)}$ and trainable parameters at $1\times$, $t = 1$; lower is better, entries are mean (seed 0, seed 1).

Predictor	MSE_1	MSE_2	MSE_3	MSE_4	MSE_5	$E^{(5)}$	Parameters
Model A	0.0168 (0.0170, 0.0167)	0.0203 (0.0207, 0.0200)	0.0226 (0.0230, 0.0221)	0.0243 (0.0248, 0.0239)	0.0259 (0.0264, 0.0255)	0.0220 (0.0224, 0.0216)	480,784
Blank	0.0426	0.0426	0.0426	0.0425	0.0425	0.0426	0
Copy-last	0.0492	0.0604	0.0669	0.0708	0.0732	0.0641	0

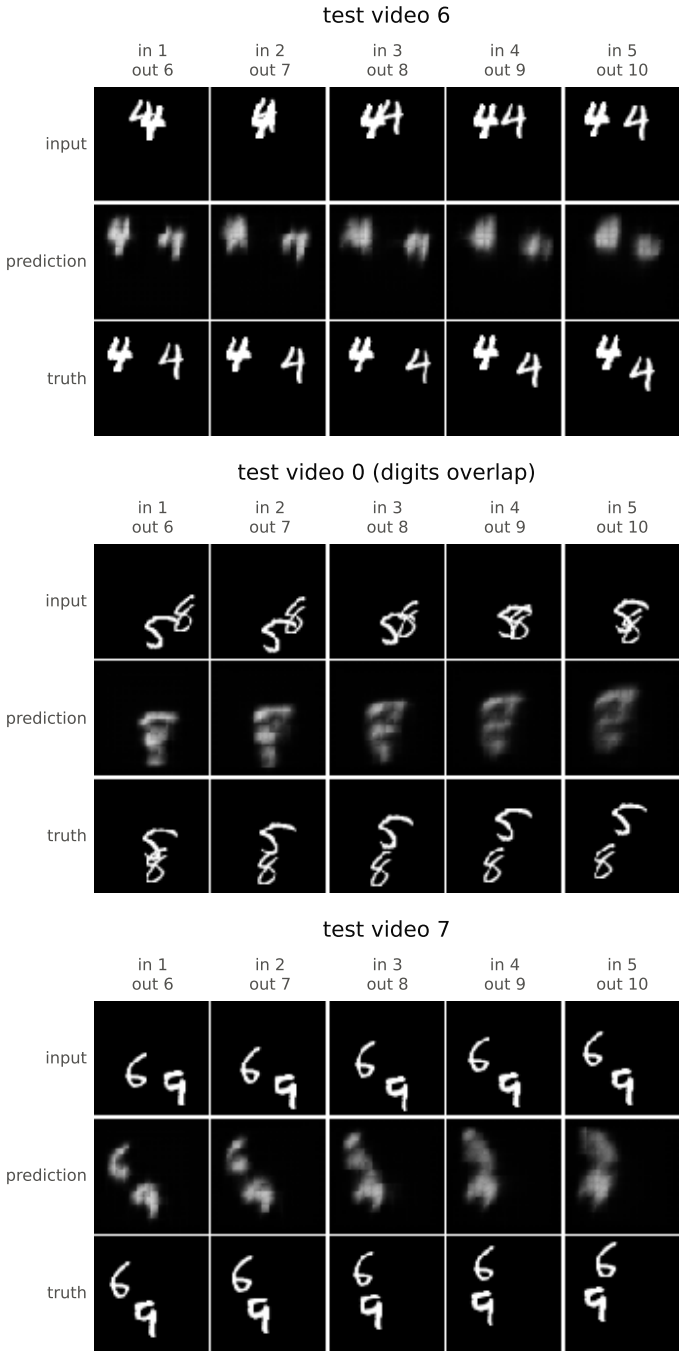


Figure 2: Model A (seed 0) on three test videos at 1×, $t = 1$. In each block: the input frames 1–5, the prediction and the ground truth for frames 6–10. Test video 0 (dataset index 9000) has overlapping digits; videos 6 and 7 (9006, 9007) do not.

Q2.3. Model A's reach is **8 pixels per time step** (2 cells of the folded grid).

- *Folding.* `pixel_unshuffle(4)` regroups each 4×4 patch into 16 channels, so one cell of the 16×16 grid is 4 px. It moves no information, and neither do the 1×1 head and `pixel_shuffle`.
- *One step τ has exactly two spatial operations:* layer 1's 3×3 convolution on $[u_\tau, h^{(1)}_{\tau-1}]$ and layer 2's 3×3 convolution on $[h^{(1)}_\tau, h^{(2)}_{\tau-1}]$. Each moves information at most one cell. The gates and the cell-state update (Eq. 4) are element-wise, so c moves nothing.
- *Paths to $\hat{x}_{\tau+1}$.*
  - From the new input: $u_\tau \to \text{conv}_1 \to \text{conv}_2$, 2 cells.
  - From layer 1's state: $h^{(1)}_{\tau-1} \to \text{conv}_1 \to \text{conv}_2$, 2 cells.
  - From layer 2's state: $h^{(2)}_{\tau-1} \to \text{conv}_2$, 1 cell.

– From layer 1's cell state: $c^{(1)}_{\tau-1} \to h^{(1)}_{\tau} \to \text{conv}_2$, 1 cell.

The longest is 2 cells, i.e. $2 \times 4 = 8$ px along each axis. Because 3×3 kernels include diagonal neighbours, it is up to $8\sqrt{2} \approx 11.3$ px diagonally.

- *Recurrence.* The recurrence does not extend this. Information can travel further over several steps, but tracking needs the digit's new position in u_τ to meet its old position, held in $h_{\tau-1}$, within one update. During the rollout the fed-back prediction plays the role of u_τ, so the same limit applies.

Q2.4.

- *Against the references.* Model A's $E^{(5)} = 0.0220$ is about half of Blank's (0.0426) and a third of Copy-last's (0.0641), and the two seeds agree closely (0.0224, 0.0216). Copy-last is worse than Blank because a stationary copy is wrong twice: it leaves ink where the digits were and none where they are now.
- *Growth over the horizons.* Model A's error rises from $\text{MSE}_1 = 0.0168$ to $\text{MSE}_5 = 0.0259$ (+54%). The first step costs most (+21%); after that it grows by 6–7% per horizon, while Blank's error stays flat. Figure 2 shows where the error comes from: positions stay roughly right, but every step loses detail, and by $h = 5$ the digits are blobs. Blurring is what a squared-error forecaster does under uncertainty, because it averages over plausible futures instead of committing to one sharp, possibly wrong, frame.
- *Overlap.* In test video 0 the 5 and the 8 overlap in the inputs, and model A predicts one merged smear that drifts with their average motion. When the true digits separate (the 5 up and right, the 8 down and left), the prediction does not split. It stretches along the line between them. Test video 7 shows the same merging as the 6 and the 9 approach each other. The ConvLSTM keeps no separate object identities: two digits on the same cells share the same state, so once they overlap their shapes and velocities mix and cannot be recovered.

Q2.5 (recorded before any Testing-the-Baseline experiment; reproduced unchanged from `report/prerun_predictions.md`). Approved by the student and committed at 2026-10-27 15:23 PKT (commit `33d4a9f`), before any of the experiments below.

> **Mirror test: should pass, approximately.** Figure 1(a) shows training motion in every direction with no empty sector, so a mirrored trajectory ($\theta \to 180^\circ - \theta$) is familiar motion. Only the digits look unfamiliar. Model A is convolutional and mostly has to carry local strokes along, so mirrored digits should cost little. Expected: E_M within about 10% of $E^{(5)}$, and C_M well below E_M.
>
> **Rotation test: should also pass, but less cleanly than mirror.** A 90° rotation ($\theta \to \theta + 90^\circ$) again gives familiar motion. But it turns vertical strokes into horizontal ones, and the learned 3×3 filters are not rotation-equivariant, so the appearance shift is larger. Expected: $E_R > E_M$ and $C_R > C_M$, with E_R up to about 20–30% above $E^{(5)}$. Degraded, not broken.
>
> **Speed-up: should break down at 3×.** Model A's reach is 8 px per time step along each axis (Q2.3; up to $8\sqrt{2} \approx 11.3$ px diagonally). The median training speed is 3.3 px/frame (Figure 1(b)), so $2\times \approx 6.6$ px/frame and $3\times \approx 10$ px/frame. At 2×, motion stays within reach but is faster than anything in training (90th percentile 4.1 px/frame). Predictions should degrade, with digits lagging behind their true positions, but stay roughly on track. At 3×, the displacement exceeds the reach for most directions (any motion within about 37° of an axis). The network cannot link a digit's old position to its new one in one update. Digits should lag, smear or fade, and $E^{(2)}$ should come close to, or exceed, Copy-last's.

2.3 Testing the Baseline

Mirror and rotation.

Table 3: Test error and consistency under mirror (M) and rotation (R) at 1×; lower is better, entries are mean (seed 0, seed 1).

Predictor	$E^{(5)}$	C_M	E_M	C_R	E_R
Model A	0.0220 (0.0224, 0.0216)	0.00186 (0.00179, 0.00194)	0.0226 (0.0231, 0.0222)	0.00312 (0.00315, 0.00309)	0.0238 (0.0241, 0.0235)
Blank	0.0426	0	0.0426	0	0.0426
Copy-last	0.0641	0	0.0641	0	0.0641

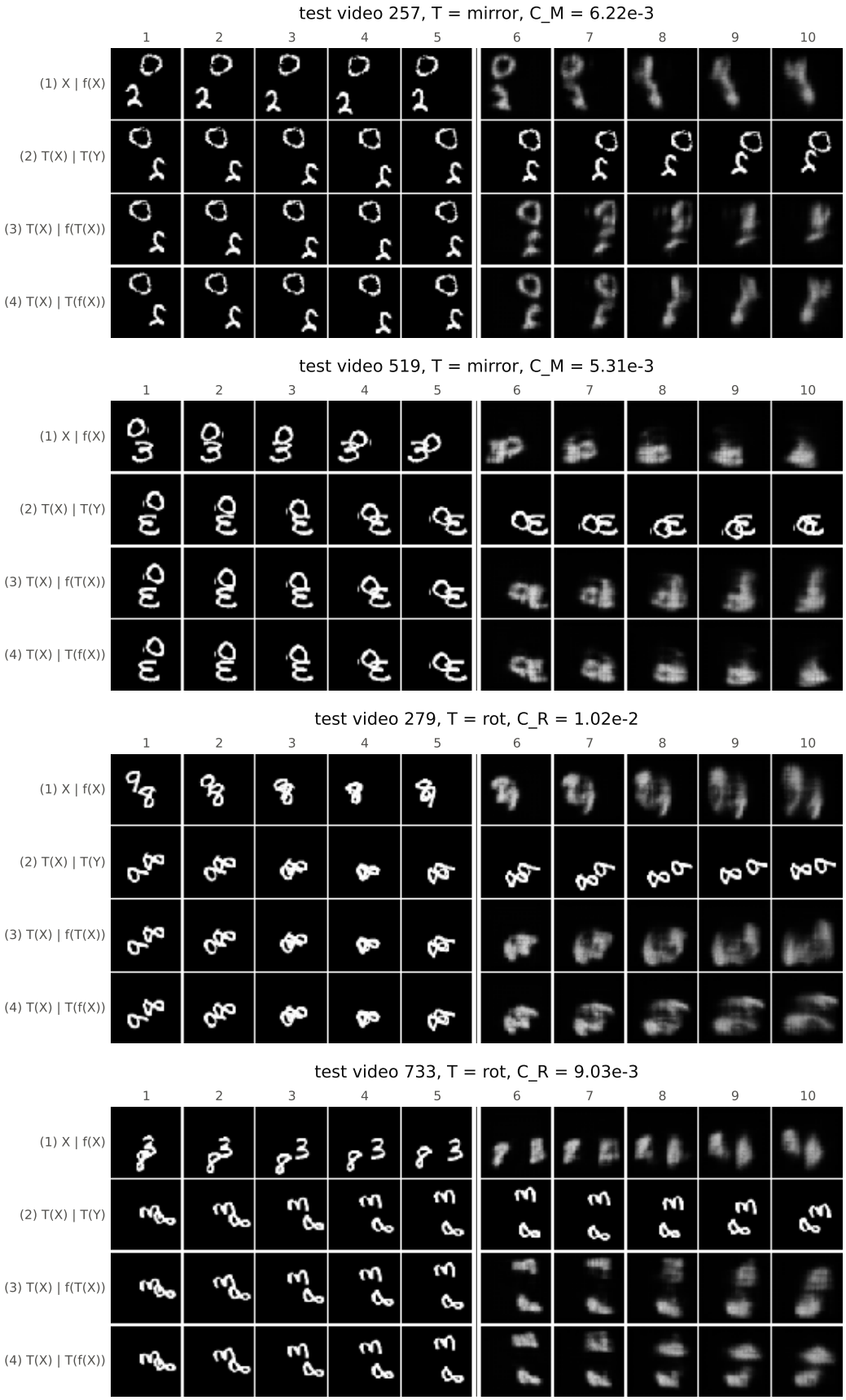


Figure 3: The two test videos with the largest per-video C_M (top two panels, T = mirror) and C_R (bottom two, T = 90° rotation) for seed-0 model A. Columns 1–5 are the inputs, columns 6–10 the five forecast horizons. Rows: (1) $X \mid f(X)$, (2) $T(X) \mid T(Y)$, (3) $T(X) \mid f(T(X))$, (4) $T(X) \mid T(f(X))$. Rows 3 and 4 should agree for a consistent model. GIFs stepping through the horizons with the same four rows: `Question2/results/gifs/fig3_mirror_257.gif`, `fig3_mirror_519.gif`, `fig3_rot_279.gif`, `fig3_rot_733.gif`.

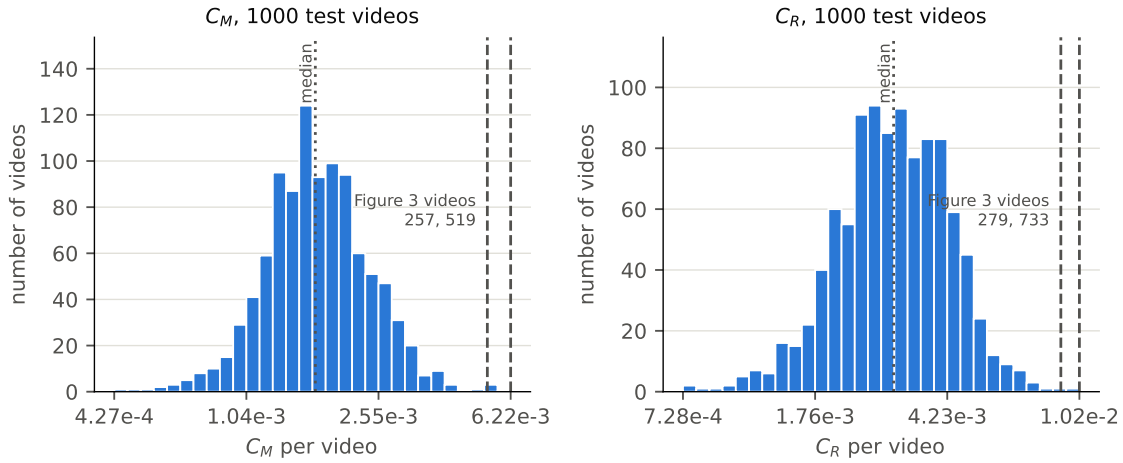


Figure 4: Per-video consistency gaps of seed-0 model A over the 1,000 test videos (log-spaced bins). Dotted line: the median. Dashed lines: the Figure 3 videos.

Q2.6. Blank and Copy-last commute with every pixel rearrangement. The zero frame stays zero, and copying the last frame and then transforming gives the same result as transforming and then copying. So $C_M = C_R = 0$ exactly, yet their transformed errors (0.0426 and 0.0641) are two to three times model A's. A forecaster that ignores the motion altogether is perfectly consistent: C_T only says whether the model treats a transformed scene the *same way*, not whether it treats it *correctly*. E_T measures the second, so a small C_T is only good news next to a small E_T. Model A has both: $C_M = 1.86 \times 10^{-3}$ is about 8% of $E_M = 0.0226$.

Q2.7. Test video 257 under mirroring ($C_M = 6.22 \times 10^{-3}$, the largest).

- *The truth* (row 2): the 0 stays near the top right, while the mirrored 2 climbs up and to the right towards it.
- *Row 4* (the upright forecast, mirrored afterwards) keeps that account. The 2 moves up towards the 0, and by frame 10 the two have merged into one leaning, forked streak near the true position.
- *Row 3* (the forecast from the mirrored input) loses the 2. It fades after frame 7, and the 0 melts into a single vertical bar below and to the left of where the digits actually go.

Shown the same motion with mirrored digits, the model no longer moves the 2 at all: one digit gets lost, and the other blurs into an unrecognisable stroke that drifts the wrong way. All four Figure 3 videos have digits that overlap or close in on each other, where the shapes are needed to tell the two digits apart.

Q2.8. Yes, rotation is harder:

- $C_R = 3.12 \times 10^{-3}$ is 1.7 times $C_M = 1.86 \times 10^{-3}$.
- E_R is 8% above $E^{(5)}$, against 3% for E_M.
- Both seeds agree, and Figure 4 shows that the whole C_R distribution is shifted to the right, not just its tail.

The motion cannot explain the difference. By Q2.1 both transformations give equally familiar motion, because the training directions cover the circle evenly and speeds are unchanged. By Q2.2, what separates them is how much they change the digits' appearance. A mirror keeps vertical strokes vertical and horizontal strokes horizontal; a 90° rotation swaps them, so rotated digits are further from anything the filters were trained on. The gap therefore comes from appearance. It is largest where two digits overlap and the model has to rely on their shapes to keep them apart.

Faster motion.

Table 4: Test error of model A at speeds $1\times, 2\times, 3\times$ (start $t = 1$); lower is better, entries are mean (seed 0, seed 1).

Predictor	$E^{(2)}$ 1×	$E^{(2)}$ 2×	$E^{(2)}$ 3×	$E^{(5)}$ 1×	$E^{(5)}$ 2×
Model A	0.0186 (0.0188, 0.0184)	0.0461 (0.0461, 0.0461)	0.0540 (0.0537, 0.0543)	0.0220 (0.0224, 0.0216)	0.0515 (0.0509, 0.0522)
Copy-last	0.0548	0.0661	0.0703	0.0641	0.0719

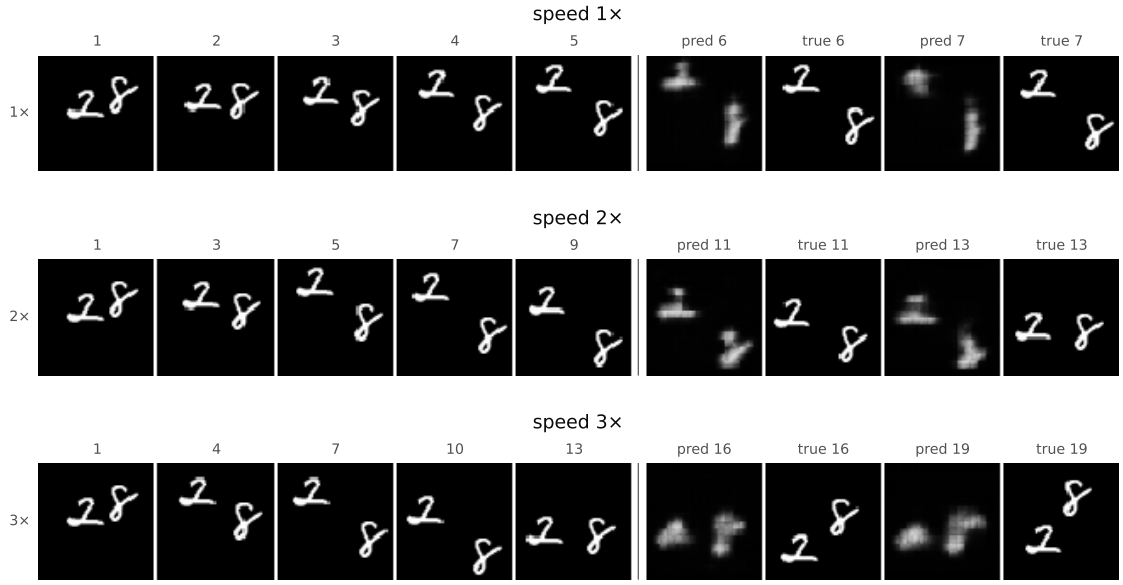


Figure 5: Seed-0 model A on test video 29 at 1×, 2× and 3× ($t = 1$): the five input frames, then the first two predicted and true future frames. The digits do not overlap in any displayed true frame.

Q2.9. Model A's predictions fall apart at **2**×:

- $E^{(2)}$ rises from 0.0186 at 1× to 0.0461 at 2× (2.5 times), already worse than predicting an empty frame (Blank: 0.0426).
- At 3× it reaches 0.0540. It stays below Copy-last (0.0661 and 0.0703) only because a blurred guess costs less than a sharp digit in the wrong place.

This only partly agrees with the reach estimate. The 3× failure fits, because about 10 px/frame exceeds the 8 px reach for most directions. The 2× failure does not, because about 6.6 px/frame is within reach. The estimate missed that reach is necessary but not sufficient. It counts the longest path that *could* carry information, but the network only learned to use the displacements it was trained on (3–4 px/frame), and the contribution from the edge of the 5×5-cell window is weak.

Figure 5 shows what happens instead: *stalling*. At 2×, the predicted 2 and 8 stay near their positions in the last input frame (9), while the true 2 moves on and the true 8 bounces upwards. At 3×, both digits sit where they were in frame 13 and fade into blobs. There are no duplicated or wrongly moving digits. The model cannot link the new positions to the old ones, so it keeps the mass where it last saw it and blurs it to hedge. This is the behaviour of a model that has not found the motion, rather than one that applies the wrong motion.

Q2.10. No. For model A the two explanations coincide: both 2× and 3× are faster than anything in training, and 3× is also beyond reach. The 2× failure shows that unfamiliar motion *alone* can break the model, which makes it impossible to decide from these runs whether reach also matters at 3×. The separating experiment keeps the architecture and changes only the data:

- Train the same network with speed augmentation ($k \sim U\{1, 2, 3\}$), i.e. model C.
- If C fixes 2× but still fails at 3×, the 3× failure is architectural reach.
- If C also fixes 3×, the failure was unfamiliar motion.
- The complementary control is a model with larger reach trained on the same speed-augmented data (model D): it should succeed at 3× exactly where C fails, if reach is the limit.

2.4 What Can Data Fix?

2.5 The Final Model

A Use of generative AI

Tool: Claude (Anthropic) via Claude Code, used throughout as a coding, analysis and writing assistant; it wrote and executed code in the student's environment and drafted report text. In line with the course AI policy, each row gives the prompt (paraphrased in formal language, preserving its substance), what the tool produced, and what the student changed or decided. Every number in this report is produced by, and reproducible from, the code in the repository.

#	Stage	Prompt (paraphrased)	Output	Student edits and decisions
1	Planning	Read the assignment handout and its front-page instructions. Without writing code or installing anything, assess whether both questions can be completed, and propose an approach that keeps accuracy while limiting cost, delegating sub-tasks to lower-cost models.	A feasibility analysis for Question 1 (written design) and Question 2 (experiment pipeline checked against the handout's built-in checks). A diagnosis of why the local GPU driver failed to load. A staged plan with a model allocation per sub-task, and a list of decisions for the student.	Reported the deadline extension to Oct 29. Fixed the GPU driver personally by enrolling the Secure Boot key. Chose to manage all GitHub commits and pushes personally. Asked for the Q2.5 and Q2.11 predictions to be drafted by the tool and reviewed by the student before they are recorded.
2	Environment	Set up the environment before any experiment, following the approach used in Assignment 1.	A Python 3.12 virtual environment reusing the existing CUDA build of PyTorch 2.9.0, with the matching torchvision 0.24.0, OpenCV, matplotlib and imageio. The Moving MNIST download. A repository skeleton (report folder, this log, the pre-run predictions file).	None.

References

- X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, W. Woo. Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting. *NeurIPS*, 2015.
- N. Srivastava, E. Mansimov, R. Salakhutdinov. Unsupervised Learning of Video Representations using LSTMs. *ICML*, 2015.